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Relationship of floral morphology and development with the pattern of bee visitation in a species with pollen-flowers, *Chamaecrista desvauxii* (Fabaceae)

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Chamaecrista exhibits a suite of floral traits associated with buzz pollination, and its morphology and organogenesis play crucial roles in pollination because bees can exploit anthers at distinct developmental stages. Here we investigated the relationship of floral morphology and development with the bee visitation pattern in *Chamaecrista desvauxii* var. *latistipula*. We also explored pollen structure, germination and stigma receptivity. Sepal growth was initially rapid and gradually slowed down as growth of the inner whorls accelerated. Heteranthery arose early during flower development and was not accompanied by morphological or histochemical differences in pollen grains, suggesting that stamen differentiation in the context of the 'division-of-labour' hypothesis was in no way associated with pollen differences. The stigma was receptive and pollen grains from both anther morphs were viable before anthesis. Bee species showed specific behaviours on anthers at both flower stages, from chewing to buzzing, in some cases touching the stigmatic region during pollen collection. Pollination experiments indicated similar fruit set between anthetic flowers and pre-anthetic buds, suggesting that pollination could occur before anthesis, even though floral buds have distinct whorl arrangements. This study highlights the existence of an extended window of opportunity for mutualistic interactions between bees and flowers in a typical species of Fabaceae with pollen-flowers.

ADDITIONAL KEYWORDS: Brazil – flower development – Leguminosae – pollen grains – pollination ecology – pollinator behaviour.

INTRODUCTION

In many bee-pollinated plants, flowers become specialized as pollen-only providers, releasing pollen grains through small pores or slits at the apex of anthers (Willmer, 2011). The limited pollen release in nectarless flowers with poricidal anthers and the occurrence of bee sonication behaviour on these anthers are considered fundamental features in the buzz-pollination syndrome (Buchmann & Hurley, 1978)

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that have evolved many times during angiosperm diversification (Vallejo-Marín *et al.*, 2010). However, some protection against pollen being entirely used as bee food is necessary to ensure that a certain amount of pollen is available for transfer among flowers. Specific floral features, such as stamen and anther dimorphism into pollinating and feeding functions by long and short stamens, respectively, would minimize the 'pollen dilemma' (Lunau *et al.*, 2015), as originally proposed by Müller (1881, 1882, 1883) in the 'division-of-labour' hypothesis. In fact, the repeated independent evolution of heteranthery associated with buzz pollination is hypothesized to be a consequence of the conflict related to the double function of pollen grains; on the one hand, they function as the only nutritional resource for pollinators, and on the other hand, they act as carriers of the male gamete for fertilization (Vallejo-Marín *et al.*, 2010).

Besides stamen and anther dimorphism, some differences may occur in pollen grains produced by different anthers in heterantherous species (Luo, Gu & Zhang, 2009). Morphological dimorphism, such as pollen size (Müller, 1981), and physiological differences, such as lower pollen viability from short stamens (Nepi, Guarnierim & Pacini, 2003; Paulino, Mansano & Prenner, 2016), have been described in heterantherous species of legumes. Because the shorter stamens are those that 'lose' their pollinating function in the context of the 'division-of-labour' hypothesis, pollen grains from those stamens could cease to be functional without compromising ovule fertilization. Indeed, the decrease of pollen viability in short stamens compared to long stamens was detected in some Fabaceae, such as *Lupinus* L. and *Cytisus scoparius* (L.) Link (Paulino *et al.*, 2016), two species of *Senna* Mill. (Luo *et al.*, 2009) and *Cassia fistula* L. (our personal observations). Based on an evolutionary timescale, division of labour could explain the occurrence of sterile pollen grains in feeding stamens (Luo, Zhang & Renner, 2008). However, morphological and functional differences of pollen grains from short and long stamens have not been considered as a stable feature, because in many species, pollen grains from both stamen types exhibit similar structure and viability (Gross & Kukuk, 2001; Luo *et al.*, 2009).

Although floral morphology in Fabaceae is diverse, most species of subtribe Cassiinae share specialized floral traits uncommon in the family. Their yellow corolla, absence of nectary, poricidal anthers, enantiostyly, heteranthery and asymmetry are floral features that are considered to have evolved several times in Cassiinae owing to convergent evolution towards buzz pollination (Gottsberger & Silberbauer-Gottsberger, 1988; Tucker, 1996, 1997; Marazzi *et al.*, 2006; Marazzi, Conti & Endress, 2007; Marazzi & Endress, 2008). This suite of floral traits favours pollination when pollen-collecting female bees vibrate flowers to access pollen grains enclosed in anthers (De Luca & Vallejo-Marín, 2013; Buchmann, 1983; Tucker, 1996). Heteranthery in Cassiinae is generally assumed to promote division of labour among stamens and, as such, it is a pollen-saving feature. However, there is a complex of different kinds of stamens and anthers that can vary among plant species and does not always reflect differences in the function of the androecium (Amorin *et al.*, 2017). Indeed, some recent studies have advanced evidence that supports the 'division-of-labour' hypothesis in species of Cassiinae (e.g. Luo *et al.*, 2009; Amorin *et al.*, 2017), but other studies have refuted the hypothesis that the two groups of anthers have distinct roles (e.g. Wolfe & Estes, 1992).

Chamaecrista (L.) Moench is a member of Cassiinae with species showing a huge morphological diversity of stamens and petals probably linked to different mechanisms of pollen transfer (Tucker, 1996; Almeida *et al.*, 2013a). Nectarless flowers and poricidal dehiscence of anthers are floral characters that place *Chamaecrista* spp. in the category of specialized pollen-only flowers (Vogel, 1978) that exclusively depend on bee sonication behaviour for pollination (Gottsberger & Silberbauer-Gottsberger, 1988). Additionally, specific morphology in combination with heteranthery in *Chamaecrista* is assumed to reduce the 'pollen dilemma' phenomenon. For example, some *Chamaecrista* spp. have a modified tube-like petal around the short stamens. This morphological feature acts as a prolongation of stamens (Gottsberger & Silberbauer-Gottsberger, 1988), much like a functional unit of one long pollinating stamen. Other species show an enlarged, rigid and hooded petal, called cucullus or kell, next to nine of the ten stamens with variable lengths, among which one single and longer stamen is deflexed and considered the pollinating stamen (Tucker, 1996) [e.g. *Chamaecrista fasciculata* (Michx.) Greene; Wolfe & Estes, 1992]. In other *Chamaecrista* spp., the cucullus is associated with all ten grouped stamens with variable lengths, such as seven or eight short feeding stamens and two or three long pollinating stamens (e.g. *C. nictitans* Moench; Tucker, 1996).

Based on their richness and abundance, *Chamaecrista* spp. are recognized as important pollen sources for bee assemblages in Neotropical savannas (Gottsberger & Silberbauer-Gottsberger, 1988). From the perspective of the animal, the anthers of such pollen-flowers increase their nutritional value during floral development owing to the process of pollen maturation (García, Nepi & Pacini, 2017). However, from the perspective of the plant, pollen must be available for collection by animals only during anthesis, which is when floral morphology and whorl

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position favour adequate deposition of pollen on the body of the bee (Almeida *et al.*, 2013b, 2015) and transfer to a conspecific stigma. In fact, germination of mature pollen is more likely to occur on receptive stigmas during anthesis, which is when the chances of fertilization are supposed to be higher. In general, therefore, floral morphology and its ontogeny play crucial roles in pollination success.

In this paper, we aimed to investigate the relationship between floral morphology and development in the context of pollen grain exploitation by bee visitors in *Chamaecrista desvauxii* (Collad.) Killip var. *latistipula* (Benth.) G.P.Lewis (Caesalpinioideae, Fabaceae), a species with pollen-flowers that are exclusively pollinated by bees (Costa *et al.*, 2007). Because *C. desvauxii* var. *latistipula* has a heteromorphic androecium, suggesting 'division-of-labour', differences in pollen features in anther types were further explored. Specifically, we asked the following questions. (1) What patterns of floral development are associated with heteranthery and pollination? (2) What differences in pollen grains produced by anthers with distinct lengths might support the 'division-of-labour' hypothesis? (3) Are female and male functions active throughout floral development? (4) What temporal pattern of bee visitation and bee behaviour might be exhibited at different floral developmental stages? (5) How are floral morphology and biology functionally linked to bee behaviour and visitation pattern?

MATERIAL AND METHODS

Focal plant species and study area

Chamaecrista is the second richest genus in Fabaceae and the ninth largest in angiosperms in Brazil, with 207 endemic species, most occurring in the Neotropical savannas (BFG, The Brazil Flora Group, 2015). *Chamaecrista desvauxii* is a shrub (Fig. 1A) widely distributed across the savannas in Brazil, and it is currently subdivided into 17 varieties based on a few morphological differences that are mostly seen on vegetative parts (Irwin & Barneby, 1982). All varieties of *C. desvauxii* have leaves subdivided into four leaflets and a petiole with one extrafloral nectary suggested to be associated with plant defence against herbivores (e.g. Baker-Méio & Marquis, 2012). Additionally, they have self-compatible pollen-flowers (i.e. nectarless flowers offering only pollen as a resource; Faegri & van der Pijl, 1979) harvested by bees through the vibration of anthers (or sonication; Gottsberger & Silberbauer-Gottsberger, 1988; Costa *et al.*, 2007). Plants have pentamerous flowers with one whorl of sepals and petals, two whorls of fertile stamens and a unicarpellate gynoecium enclosing the ovules (Fig. 1B). The dehiscent fruits open explosively to disperse seeds.

To study flower morphology and development in the context of the 'division-of-labour' hypothesis and bee visitation pattern, we selected individual plants of *C. desvauxii* var. *latistipula*, a plant that is abundant in the city of Botucatu, São Paulo, Brazil. *Chamaechrista desvauxii* var. *latistipula* is endemic to Brazil and widely distributed from the northern to southern regions. Plants of *C. desvauxii* var. *latistipula* can be recognized from other varieties by their long pedicels (22–50 mm length), large flowers (*c.* 14–25 mm) and persistent large stipules with rounded apex (15–26 × 10–20 mm) [Irwin & Barneby (1982) named this taxon *C. desvauxii* var. *glauca* (Hassl.) Irwin & Barneby]. The individuals were sampled (vouchers BOTU 32753 and BOTU 32754) and marked in a 10-ha region around São Paulo State University (Unesp) (22°53′S, 48°29′W) in fragmented areas of open physiognomies of savanna highly disturbed by intense anthropogenic activities and surrounded by areas with invasive grass species [e.g. *Urochloa decumbens* (Stapf) R.D.Webster; Zenni & Ziller, 2011]. This region has a seasonal climate with rainy summers between November and February and dry winters between May and September. Temperatures range between 23.1 °C in the hottest month (February) and 17.1 °C in the coldest month (July), and the mean annual rainfall is *c.* 1456 mm (Cunha & Martins, 2009).

Floral morphology and inner whorl development

To investigate the changes in floral morphology during development in *C. desvauxii* var. *latistipula*, especially for pollination organs and petals (i.e. inner whorls), we selected ten shrubs to describe variability in floral morphology within the population. Because pollen resources can be exploited by bees before anthesis (our previous personal obsservations), the morphological study of floral changes in the inner whorls, including morphometrics, is fundamental to a full understanding of pollen exploitation and its consequences for sexual reproduction in this plant species.

Only mature shrubs with more than three stems growing from the same insertion point on the ground and separated at least by 5 m from each other were tagged. Currently, no evidence of clonality has been reported in this species. We randomly selected one reproductive branch per plant, including floral buds with different sizes, to describe changes in size (growth) of each floral whorl during development. This was accomplished by measuring the total floral length and the length of each inner floral whorl. We fixed at least ten floral buds per branch, including

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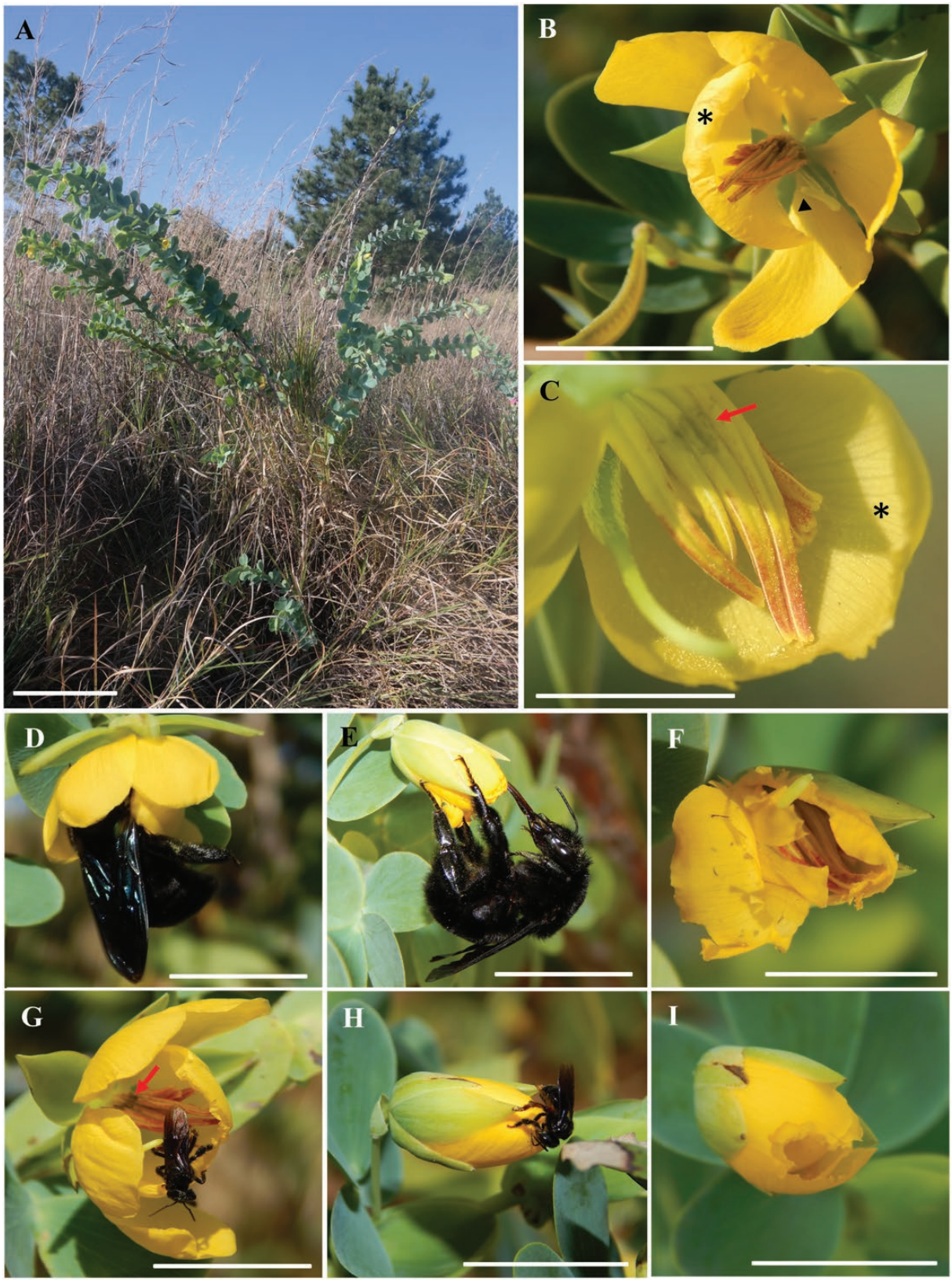


Figure 1. *Chamaecrista desvauxii* var. *latistipula*, floral morphology and bee visitors on pre-anthetic floral buds and anthetic flowers. A, *C. desvauxii* shrub in the flowering period. B, anthetic flower showing the outermost lower petal, two

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the continuous variation of bud size to open flowers, in 50% FAA (formaldehyde–acetic acid–alcohol; Johansen, 1940) and stored them in 70% ethanol. Only flowers without evidence of herbivory were analysed, resulting in the exclusion of < 3% of flowers sampled. First, we externally measured flower length in floral buds > 1 mm in length from the floral receptacle base to the apices of sepals, or petals in pre-anthetic buds. We then used a Leica stereomicroscope coupled with a camera to take sequential pictures of each floral whorl for each bud sampled from the reproductive branch. Using ImageJ software, the length of each floral whorl was measured, as well as width in the case of an inner modified petal, known as the cucullus, which grows more laterally (Rasband, 1997). Opened flower length was estimated based on the length of the longest petal. Floral whorl size was modelled in relation to flower length (in millimetres) from bud stages to flower opening.

We also used scanning electron microscopy (SEM) to perform morphological analyses of floral buds, focusing on the development of the inner floral whorls. For this, five additional plants were selected, and one reproductive branch per plant was collected to sample floral buds, including the continuous variation of bud size to open flowers. Reproductive branches were previously protected with organza bags to avoid the potential effects of herbivores or pollen extraction by pollinators. Floral buds were then fixed in 2.5% glutaraldehyde buffered with 0.067 M sodium phosphate, pH 7.2 (Gabriel, 1982), and placed under vacuum before storage at 4 °C. For SEM, fixed flowers and floral buds were further dehydrated in an ethanol series, dried in a Leica EM CPD300 drier system, mounted on aluminium stubs and coated with gold in a Leica EM ACE200 sputtering system. The images were obtained with an FEI Quanta 25 scanning electron microscope at 10 kV.

Comparison of pollen grains from long and short anthers: structure, histochemistry and germination

Because structural differentiation in pollen grains has been suggested as a reflection of the 'division-of-labour' hypothesis in pollination (see Introduction), we performed pollen grain comparisons between short and long anthers, as applied by Luo *et al.* (2009). For this, we collected new reproductive branches of 15 plants isolated in organza bags. We dissected the androecium and separated the long and short anthers for pollen counting, anatomical and histochemical analyses of pollen grains, and estimation of germination rates.

For pollen counting, we dried the long and short anthers of each flower (N = 10 flowers; 10 plants) at 50 °C. We then carefully broke the seven short anthers and three long anthers into different small Petri dishes using razor blades, and then released the pollen grains from each set of poricidal anthers. We diluted the pollen grains in 1 mL of water and homogenized the solutions. Finally, we extracted three subsamples of each solution with 5-µL microcaps and counted the pollen grains linearly using a stereomicroscope at 8× magnification.

For the anatomical and histochemical analyses of pollen grains, long and short anthers (N = 8 flowers) were dehydrated in an ethanol series, embedded in historesin Technovit 7100 (Heraeus Kulzer, Hanay, Germany) and sectioned at 1.5 µm thickness. Slides were stained with toluidine blue O (pH 6.9) (O'Brien, Feder & McCully, 1964) for structural analysis by light microscopy. Histochemical tests were performed: periodic acid-Schiff's reaction (PAS) to detect polysaccharides (McManus, 1948), xylidine Ponceau for proteins (Wetzel, Demmers & Greenwood, 1989) and copper acetate/rubeanic acid for fatty acids (Ganter & Jollés, 1969). Polyphenols were identified in fresh pollen grains by their fluorescence under UV light. Slides were mounted in water, and sections were viewed and digitally photographed with a Nikon Eclipse Ti-E compound microscope equipped with Nikon NIS Elements software.

To prepare pollen germination slides, long and short anthers (N = 22 flowers) were measured and isolated in different Petri dishes for each flower. Pollen grains from fresh flowers were then removed from the anthers and immediately placed in the germination medium prepared over microscope slides. For this, two media were prepared. The first medium was composed of 12.5% sucrose, 0.01% boric acid, 1 mM $CaCl_2$, 1 mM $Ca(NO_3)_2$, 1 mM $MgSO_4$ and 0.5% agar, a protocol usually applied to germinate pollen of *Arabidopsis* Heynh. in our laboratory. The second medium contained 25% sucrose and 1% agar, a

upper petals and two lateral petals; the large and concave innermost lateral petal on the left side is the cucullus with the black asterisk; the stigma is marked with a black triangle. C, detail of an anthetic flower after *Bombus* vibration; note many pollen grains scattered on the cucullus (black asterisk) and an injury on the base of the anthers made by *Bombus* sp. indicated by the red arrow. D, *Bombus* sp. exploiting anthers by sonication in an anthetic flower. E, *Bombus* sp. exploiting a pre-anthetic bud. F, pre-anthetic bud after *Bombus* sp. visits. G, *Trigona* sp. exploiting anthers with its mandibles in an anthetic flower. H, *Trigona* sp. exploiting a pre-anthetic bud. I, pre-anthetic bud after *Trigona* sp. visits. Flowers are orientated in their natural position in all pictures. Scale bars: 20 cm (A), 1.5 cm (B), 1 cm (C), 2 cm (D–I).

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protocol usually applied to germinate pollen of native species (Cardoso-Gustavson, Demarco & Carmello-Guerreiro, 2011). Both media exhibited similar results for pollen germination of *C. desvauxii* var. *latistipula* (data not shown), but we presented only results based on the second protocol. We prepared 22 slides with pollen grains from long anthers and 22 slides with short anthers (N = 22 flowers from five plants). Slides were kept at room temperature overnight and viewed under UV light or by brightfield microscopy, followed by digital photography with a Nikon Eclipse Ti-E compound microscope equipped with Nikon NIS Elements software.

To estimate germination of the pollen grains, microscope slides were viewed at 25× magnification. For each slide, pollen grains from three to five randomly taken microscope fields were imaged and analysed. From each field, we counted the number of pollen grains, including those that were germinated and plasmolysed. Pollen grain sizes were estimated by their total area and not by one-dimensional measurements, such as pollen length. In equatorial view, germinated pollen grains exhibited a larger area, owing to their turgid state, whereas plasmolysed pollen grains showed a smaller area compared to other pollen grains. Thus, germinated and plasmolysed pollen grains were excluded from pollen size measurements. Thus, the count of pollen grains, germinations, contractions of protoplast (plasmolysis) and pollen size were quantified using ImageJ software in each imaged microscope field. For each pollen descriptor, we calculated the average microscope field for each slide.

Female and male functions throughout floral development: pollen viability and stigma receptivity

To evaluate when female and male functions are active during flower development, we selected new reproductive branches in five plants, also isolated in organza bags. Flower length was measured in all flowers, as previously described, and we sampled at least four flowers per branch in each of the following stages: flowers at anthesis (N = 10 flowers), pre-anthetic buds (N = 6 buds) and young floral buds with intermediate lengths (N = 6 buds, Table 1), totalling 22 units. Pollen germination was checked exactly as described above. Carpels from 22 new fresh flowers at different developmental stages were carefully excised using the same five plants and the same procedures adopted in the last section (from organza-bagged branches). Their stigmas were then exposed to 3% hydrogen peroxide (peroxidase reaction) by immersion for 3 min and considered receptive when producing more than ten successive oxygen bubbles (Galen & Plowright, 1987). We also investigated trichome position occurring close to the stigmatic region in each flower,

Table 1. Floral developmental stages and pollen descriptors from different anther types of *Chamaecrista desvauxii* var. *latistipula* flowers

Floral development stages (Figures)	Days before anthesis	$N_{(flowers)}$	Flower length (mm)	Anther type	Number of anthers	Anther length (mm)	Number of pollen grains per anther	Quantitative pollen descriptors		
								Size (μm²)	Germinated (%)	Plasmolysed (%)
Anthesis (Fig. 1B–D, G)	–	10	21.4 ± 0.8	Long	3	12.1 ± 0.3	2050 ± 636	735.2 ± 100.2	25.3 ± 24.0	26.3 ± 11.1
				Short	7	8.0 ± 0.1	1090 ± 241	678.2 ± 147.0	26.9 ± 19.3	20.8 ± 9.5
Pre-anthetic buds (Fig. 1E, F, H, I)	1–3	10	20.5 ± 0.6	Long	3	12.1 ± 0.2	–	679.4 ± 120.6	6.9 ± 4.2	17.0 ± 5.9
				Short	7	7.4 ± 0.4	–	767.9 ± 109.1	7.9 ± 8.8	20.6 ± 10.9
Larger buds	5–7	10	17.8 ± 1.4	Long	3	10.3 ± 0.5	–	691.0 ± 93.3	4.5 ± 3.4	23.5 ± 3.8
				Short	7	7.2 ± 0.6	–	706.6 ± 124.9	3.2 ± 2.6	21.5 ± 10.7
Intermediate buds (Fig. 2C)	16–20	20	11.5 ± 1.1	Long	3	5.9 ± 1.1	–	–	–	–
				Short	7	4.1 ± 1.3	–	–	–	–
Tiny buds	> 30	13	4.4 ± 0.8	Long*	2	2.0 ± 1.2	–	–	–	–
				Short*	8	1.1 ± 0.6	–	–	–	–

Pollen size is described based on area (μm²); *tiny buds had only two longer anthers. Pre-anthetic buds are those that appear immediately, i.e. 1–3 days, prior to anthesis.

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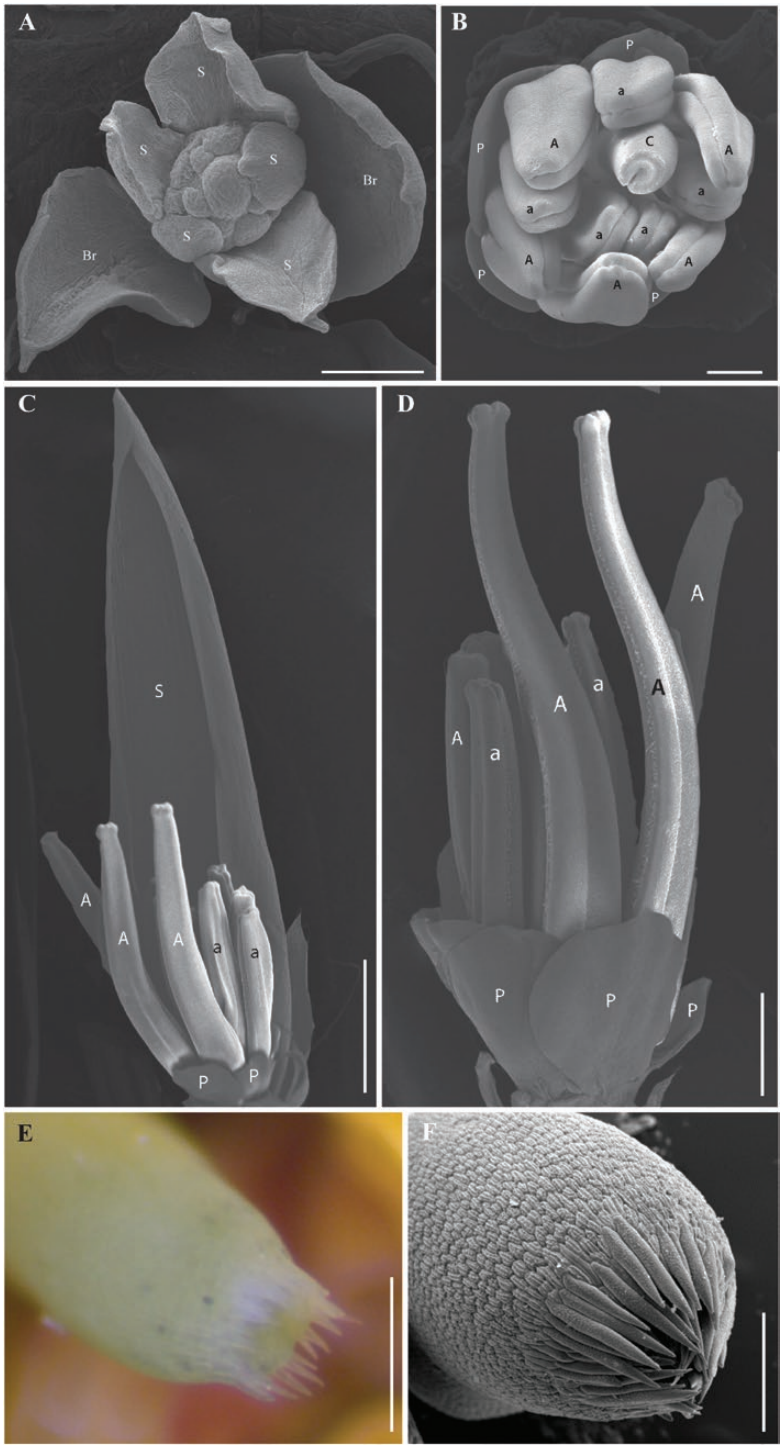


Figure 2. Floral bud morphology during development of *Chamaecrista desvauxii* var. *latistipula* flowers. A, young floral bud with two outer prophylls (*Br*) and sepals (*S*). B, later stage than in A, showing the four young inner whorls: petal whorl (*P*) delayed in development in relation to the two stamen whorls (outer whorl, *A*, with larger young stamens; inner younger whorl, *a*), and the young carpel (*C*). C, later stage than in B; floral bud with intermediate size (around 9 mm) still with delayed petal growth. D, later stage than in C, after floral bud elongation phase (around 14 mm), with petals larger and starting their exponential growth. E, stigma morphology of a fresh anthetic virgin flower. F, SEM micrograph of the stigma in an anthetic virgin flower. Scale bars: 0.2 mm (A, B), 2 mm (C), 1 mm (D, E), 400 μm (F).

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using a stereomicroscope, because other *Chamaecrista* spp. are reported to have a fringe of hairs obscuring the stigmatic surface (Owens, 1989; Arceo-Gómez *et al.*, 2011).

Bee visitation pattern at different floral developmental stages

Because floral bee visitors differ in their rate of visitation on flowers throughout the day and among species, we initially sought to describe the temporal pattern of bee visits. As pollen is the only floral resource of *C. desvauxii* var. *latistipula* and flower opening occurs early in the morning, our expectation was that bee visits would be more abundant in the early morning and then gradually diminish as the pollen was extracted from poricidal anthers. To test this assumption, we systematically assigned 35 plants across the entire population of *C. desvauxii* var. *latistipula* (*c.* 12 000 m^2) and performed focal observations at 1-h intervals per plant to track floral visits. The temporal pattern of bee visitation and behaviour during visitation were recorded for a period corresponding to the entire flower lifespan after anthesis (between 06:00 and 12:00 h), totalling 78 h of observations. We recorded the number of anthetic flowers and pre-anthetic buds visited by each bee species (per plant) and the number of anther vibrations made by each floral visitor during pollen collection, as determined by discrete buzzes noticed during each visit. For taxonomic identification of bee visitors, we collected a few individuals of each bee species. Subsequently, we checked our identifications using the bee collection of Dr Guimarães, Department of Botany, São Paulo State University (Unesp), Botucatu.

We also qualitatively estimated the amount of pollen grains removed from anthers after successive bee visits at the end of anthesis. To do this, we sampled fresh flowers at noon, longitudinally cut each anther and, with the aid of a stereomicroscope, visually quantified how many anther locules were empty after successive visits based on the percentage of pollen grains removed, i.e. 0–24, 25–49, 50–90 and > 90%.

Manual pollination experiments

Experimental crosses were performed in the field to evaluate the capacity of pre-anthetic buds and anthetic flowers to produce fruits. This procedure was adopted to simulate the effect of potential visits of pollinating bees at different moments of floral development. In this experiment, ten plants were selected in the field, and at least three reproductive branches per plant were previously isolated in organza bags. We carried out four different treatments, one flower per plant per treatment: (1) anthetic flowers receiving pollen grains from anthetic flowers, (2) pre-anthetic buds receiving pollen grains from anthetic flowers, (3) pre-anthetic buds receiving pollen grains from pre-anthetic buds and (4) pre-anthetic buds completely excluded from bee visitors and pollen transfer and thus subject to spontaneous self-pollination treatment. In this last treatment, the absence of fruit and seed formation could be interpreted as the absence of autonomous self-pollination and any kind of apomixis. In the first three treatments of experimental pollen transfer, we used only outcrossed pollen grains removed from anthers using a small toothpick, immediately transferring the grains to the stigma of the recipient flower. After pollen transfer, we bagged all reproductive branches again and verified fruit formation per flower over time.

Statistical analyses

To describe the changes in the inner whorls during flower development, we modelled the size of each inner floral whorl, including lower petal length, cucullus width, and stamen and carpel length, by the total length of floral buds, using two alternative functions, exponential and linear, in general mixed models (LMMs) with Gaussian error distribution. In those cases, the length of each floral whorl was analysed as the response variable, flower length, including bud length as the continuous fixed factor and individual plants as the random variable.

We also investigated the 'division-of-labour' hypothesis based on functional and morphological descriptors of pollen grains from long and short anthers. For this, we applied paired *t*-tests to determine any differences in the number of pollen grains per anther, pollen size (μm^2) and percentage of germinated and plasmolysed pollen grains between long and short anthers in anthetic flowers. We also evaluated differences in pollen germination between anther types throughout flower development. Male function by flower length was evaluated by a generalized linear mixed modelling (GLMM), as discussed below.

To investigate male and female function in relation to flower size, we modelled the proportion of germinated pollen grains and stigma receptivity among flowers with different lengths. For male function, we used GLMM with a binomial error distribution, in which flower length and anther type were included as continuous and categorical fixed factors, respectively, and individual plants were considered as the random variable. Concomitantly, to investigate female function, we separately modelled the probability of the stigma becoming receptive by flower length using GLMM with a binomial error distribution.

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To explore differences in bee visitation patterns between anthetic flowers and pre-anthetic buds, we first applied a binomial proportion test to compare differences in the proportions of visited plants with anthetic flowers and pre-anthetic buds. In a complementary manner, we tested for differences in the number of bee visits per hour per plant, as well as between anthetic flowers and pre-anthetic buds, independent of bee identity, using GLMM with Poisson error distribution, considering bee visits as the fixed factor (counting variable) and individual plants as the random variable.

Bee visitation pattern throughout the day was modelled separately for species of *Bombus*, *Oxaea* and *Trigona*, because these bees differ in their visitation pattern through time, mainly resulting from variations in floral resource availability. Although we expected bee visits to be more frequent in the early morning before the gradual decrease of pollen grain availability, we did not have any prior reason to choose a particular parametric shape to describe the relationship between bee visits and time of day. Therefore, we applied a generalized additive model (GAM) for each bee species. This strategy modelled time of day as non-parametric smoothed function s to explain the number of flowers visited per hour (counting variable). A zero inflated Poisson error distribution was used in these GAMs.

After each analysis using generalized models, we checked for the occurrence of over-dispersion in the dataset before interpreting the significance of each effect. We used the ratio between the residual deviance and the residual degrees of freedom, or scale parameter, to describe the over-dispersion that occurs when the observed variance is greater than the variance of the theoretical model. Assuming an appropriate error distribution in each model, both residuals would have similar values, and the scale parameter would be around 1 with an excellent best-fit (Crawley, 2007). In the general linear models, we also checked for the occurrence of normal distribution and homogeneity of residuals. All statistical analyses were carried out in R v. 3.1.1 (R Development Core Team, 2017) with standard and additional packages: mgcv (Wood, 2001) and glmmADMB (Bolker *et al.*, 2012).

RESULTS

Floral morphology and inner whorl development

At the beginning of flower development, two prophylls subtend each individual flower, thereby protecting the young floral meristem (Fig. 2A). The calyx then assumes the protective function of the inner floral whorls in floral buds of intermediate size (Fig. 2C, Table 1) until the pre-anthetic phase when this function is assumed by the expanded corolla (Fig. 1E, H; Table 1). Reproductive plants had one to 13 opened flowers per day. Anthetic flowers are asymmetric, monomorphic and enantiostylous, arranged in short few-flowered inflorescences, a reduced axillary raceme generally with a single flower (range: one to three flowers per raceme). On average, 49.5 and 50.5% of flowers are right- and left-styled, respectively (N = 30 plants), although, in general, plants with fewer than three simultaneously anthetic flowers had only one floral morph at anthesis at the same time.

The fully developed corolla has one outer lower petal, one lateral petal, two upper petals and one innermost lateral petal enlarged, concave and slightly thicker, which is called the cucullus (Fig. 1B). The heterantherous androecium is diplostemonous with ten fertile stamens, three long and seven short, close together as one group and deflected towards the inner side of the cucullus in anthetic flowers (Fig. 1B, C). Anthers are poricidal with two pores at their distal region. In contrast, the single carpel is placed sideways to the cucullus, almost opposite to it, with the style and stigma deflected toward the inner side of the cucullus in anthetic flowers (Fig. 1B). The mature stigma is crateriform with its entrance porate filled with visible exudate (wet stigma) and lined with short erect trichomes. The stigmatic trichomes are erect throughout anthesis in fresh flowers (Fig. 2E; N = 30 flowers analysed). However, in flowers treated for SEM, trichomes were bent towards the centre of the stigma, closing the stigma surface (Fig. 2F).

Floral whorls have distinct patterns of growth during development. Petals (Fig. 3A: Z = 20.6, P < 0.001, N = 88; Fig. 3C: Z = 22.1, P < 0.001, N = 83) and carpel (Fig. 3D: Z = 17.6, P < 0.001, N = 83) experience delayed development in an exponential pattern, whereas the stamens show a linear growth pattern, including both morphs (Fig. 3B: Z = 22.8, P < 0.001, N = 174) in a manner similar to that of sepals. In buds < 4 mm in length, all stamens are more or less the same size (Fig. 2B). Thereafter, anthers of the two outer stamens become longer (tiny buds in Table 1), followed by elongation of the third outer stamen (Fig. 2C; Table 1). Thus, heteranthery appears early in flower development. Anthers dehisce before flower opening, although, in general, only two long anthers have opened pores in pre-anthetic buds. At this stage, stamens are similar in size to those of anthetic flowers (Table 1) with the cucullus completely enclosing the pollination organs, even though the width and the length of petals, including the cucullus, are highly variable among pre-anthetic buds (Fig. 3A, C). In fact, large interfloral variability was observed for petal and carpel sizes in flowers longer than 15 mm (Fig. 3A, C, D), although this variability was much smaller for

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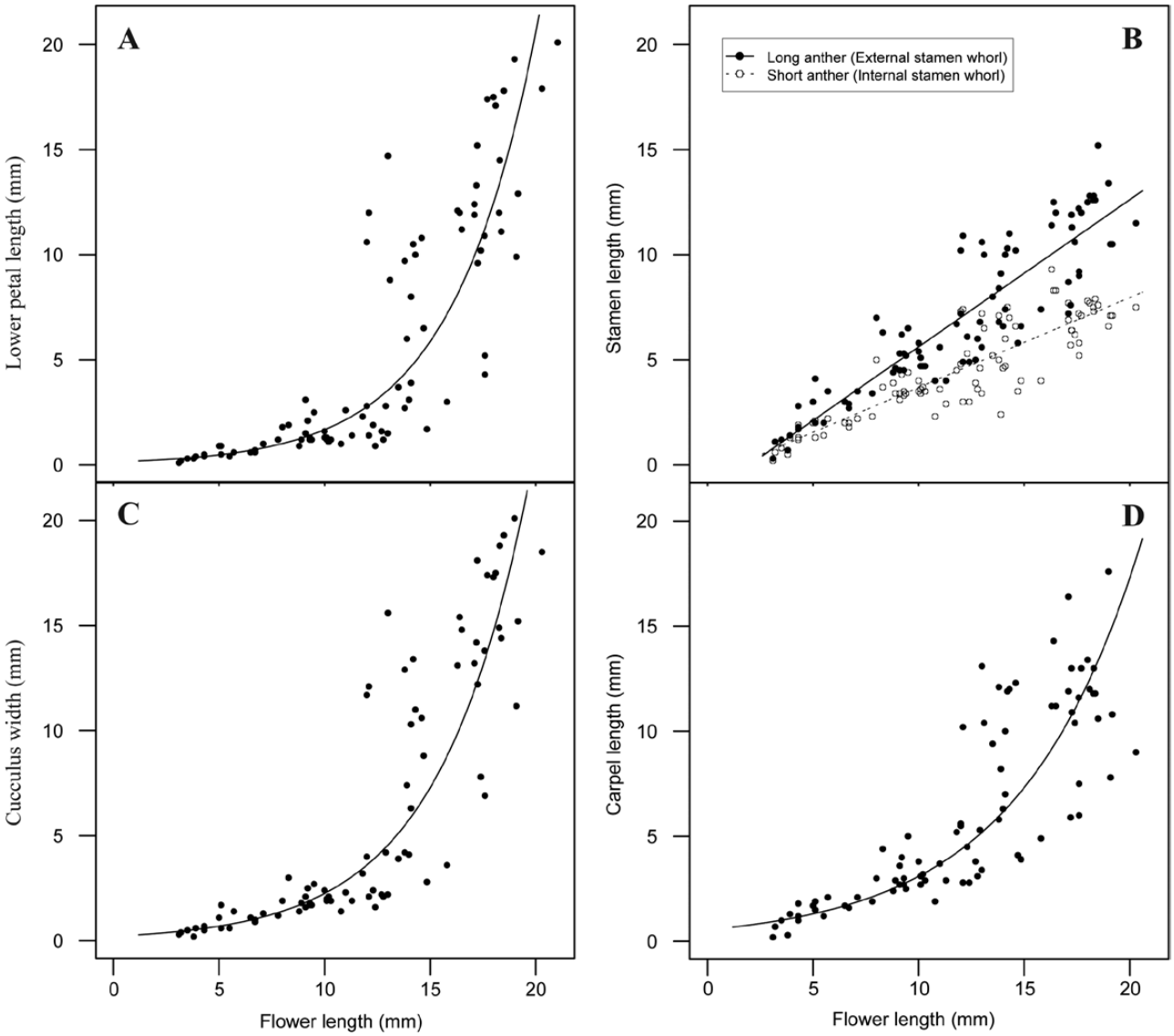


Figure 3. Relationship between flower whorl sizes and entire length of flowers from bud stage to flower opening in *Chamaecrista desvauxii* var. *latistipula*. Outermost lower petal length (A), cucullus width (C) and carpel length (D) were exponentially related to floral bud size. Long and short anthers were linearly related to floral bud size (B). The relative increase of growth of all inner whorls (petals, stamens and carpels) occurred after bud elongation phase in buds > 10 mm in length.

stamen size. Detailed statistical results can be found in the Supporting Information, Table S1.

Comparison of pollen grains from long and short anthers: structure, histochemistry and germination

Long anthers had twice as many pollen grains as short anthers (Table 1, $t_{paired} = 5.5$, $P < 0.001$). Pollen grain morphology, from both long and short anthers, was similar, and pollen grains could be classified as isopolar and tricolporate with radial symmetry (Fig. 4A, B). In anthetic flowers, pollen grains had the same size (Table 1, $t_{paired} = 1.5$, $P = 0.21$), percentage of germinated pollen grains (Table 1, $t_{paired} = 0.3$, $P = 0.74$), percentage of plasmolysed pollen grains (Table 1, $t_{paired} = 0.9$, $P = 0.42$) and histochemical features, irrespective of their anther morph origin. During flower development, pollen grains from both anther types had similarly low germination rates (Fig. 5A, $Z = 0.07$, $P = 0.07$, $N = 40$). Phenolic compounds constituted the pollen grain coating and also occurred in the vacuole of the pollen

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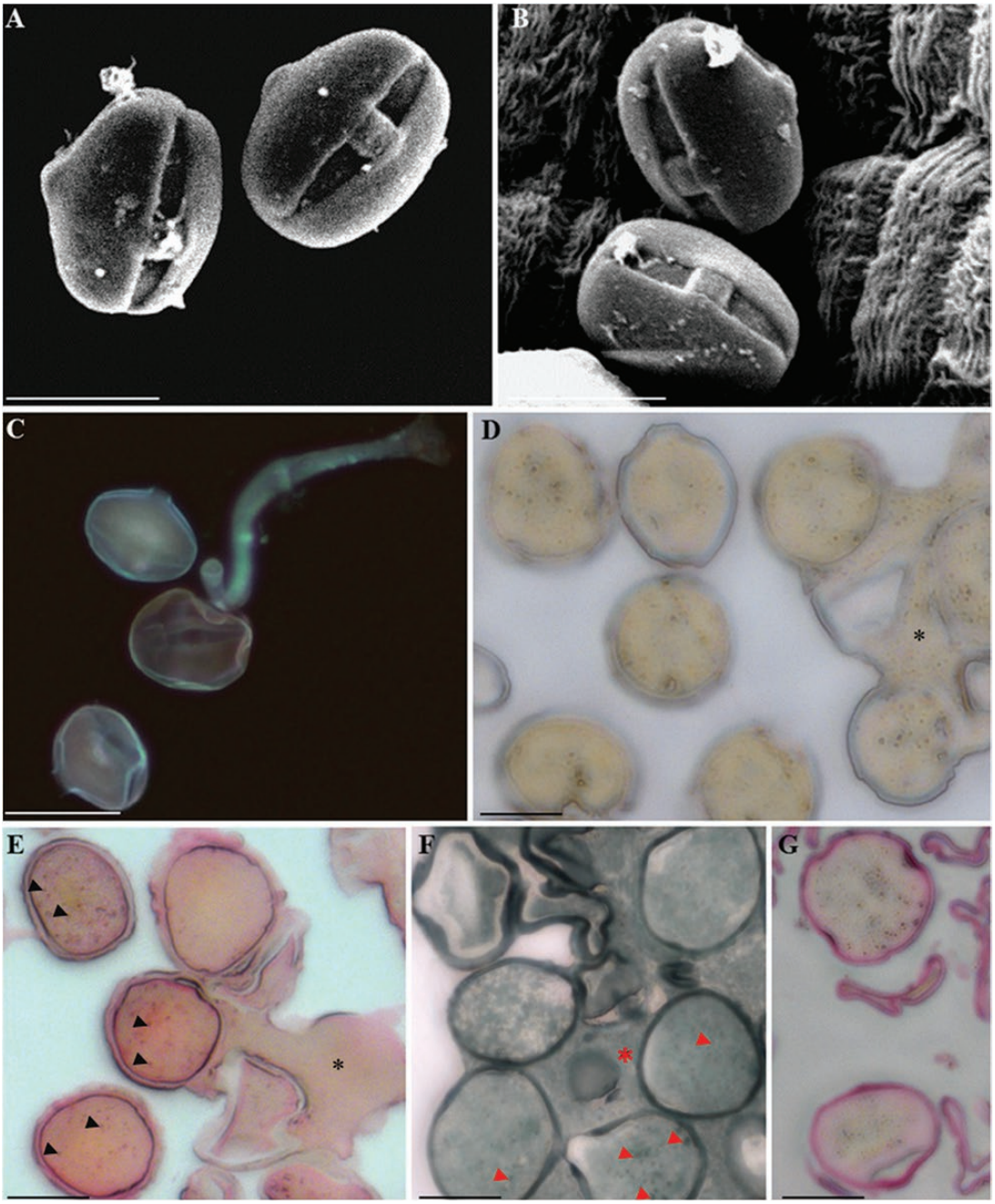


Figure 4. Shape, size and histochemistry of pollen grains from short and long anthers of *Chamaecrista desvauxii* var. *latistipula*. Equatorial view of pollen grains (SEM) from short (A) and long (B) anthers. Note the similar shape and size among pollen grains from both anther morphs. Phenolic compounds in the pollen grain coating (C), protoplast (D) and in the secretion around them (* in D). Protein bodies (black arrowheads) inside pollen grains and in the secretion around them (* in E). Fatty acids composing the lipid droplets (red arrowheads) inside the protoplast of pollen grains and in the extracellular secretion (* in F). Polysaccharides exclusively constituting the cell wall of the pollen grains (G). Scale bars: 50 μm (A, B), 10 μm (C–G).

grains, as visualized by a slightly brown coloration in sections without further staining (Fig. 4C, D). Protein bodies (Fig. 4E) and mainly lipid droplets (Fig. 4F) were identified as storage compounds, but starch grains were not detected (Fig. 4G). In addition, phenolic compounds, proteins and fatty acids constituted the exudates observed around the pollen grains (pollenkitt) (Fig. 4D–F).

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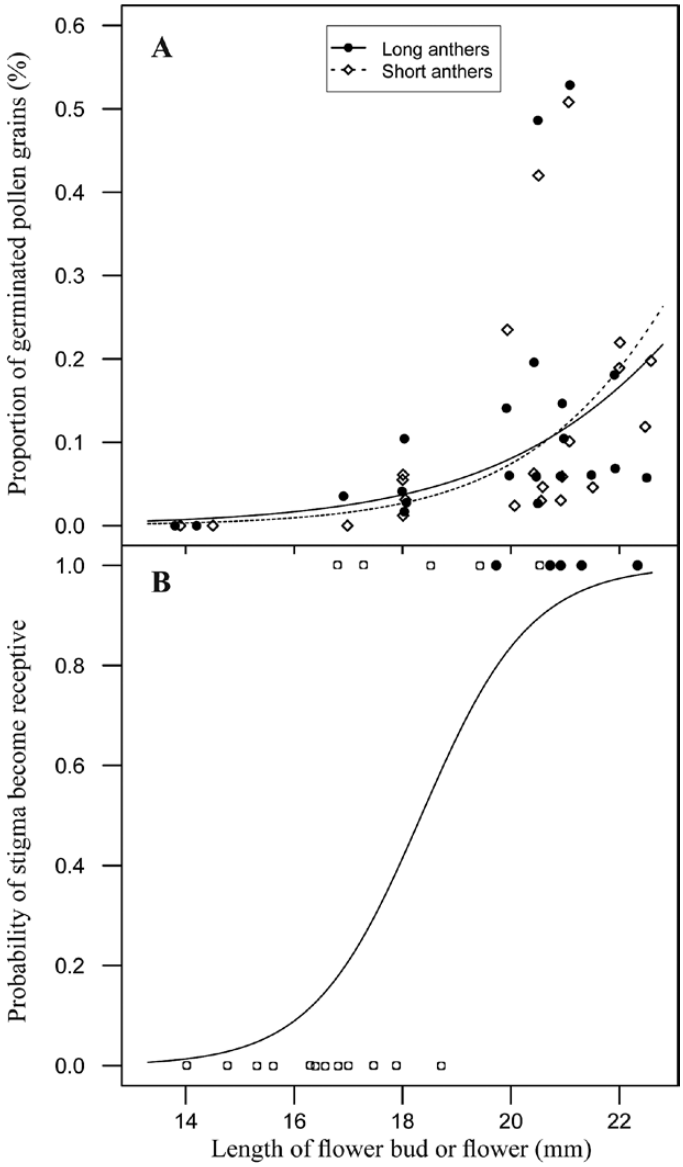


Figure 5. Female and male function during flower development of *C. desvauxii* var. *latistipula*. A, Pollen grains have lower germination rate during bud development, including the period of flower opening. Long and short anthers have similar pollen germination rates during flower development. B, Stigma receptivity is more likely when flowers reach more than 18 mm in length, a floral size near that of the flower opening period (black circles).

Female and male functions throughout floral development: pollen viability and stigma receptivity

Flowers of *C. desvauxii* var. *latistipula* displayed viable pollen grains and a receptive stigma before flower opening (Fig. 5; Table S3). The highest pollen germination proportion (*c*. 20%) was observed only in pre-anthetic buds and immediately after flower opening (Fig. 5A, $Z = 11.8$, $P < 0.001$, $N = 40$). The stigma of floral buds > 18 mm in length had a better than 50% chance of being receptive (Fig. 5B, $Z = 2.4$, $P = 0.02$, $N = 20$). Detailed statistical results can be found in the Supporting Information, Table S2.

Bee visitation pattern at different floral developmental stages

Most flowers analysed (*c*. 90%) opened between 06:00 and 07:00 h, and others opened in the next few hours. On average, each sampled plant produced 5.0 ± 2.9 flowers per day (mean ± SD, $N = 36$ plants). Opened flowers remained fresh on the plants until midday,

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after which flowers began to wither. Plants exhibited opened flowers and floral buds simultaneously. Pollen grains in anthers of anthetic flowers and pre-anthetic buds were both exploited by bees (Fig. 1D, E, G, H); however, the proportion of plants with opened flowers visited by bees was higher than the proportion of plants with only pre-anthetic buds (Fig. 6A). Additionally, anthetic flowers received, on average, four times more visits than pre-anthetic floral buds (Fig. 6B). Bees visited each anthetic flower four or five times throughout anthesis, on average, and up to ten times. Flowers visited successive times by bees throughout the morning without anther damage showed a decrease in the amount of pollen grains by > 90% by the end of anthesis.

We observed five different bee species on flowers of *C. desvauxii* var. *latistipula*. The most common visitors were *Bombus* sp. (51% of visits), *Trigona* sp. (42% of visits) and *Oxaea* sp. (4% of visits). Bee visits began after dawn around 07:00 h and continued until

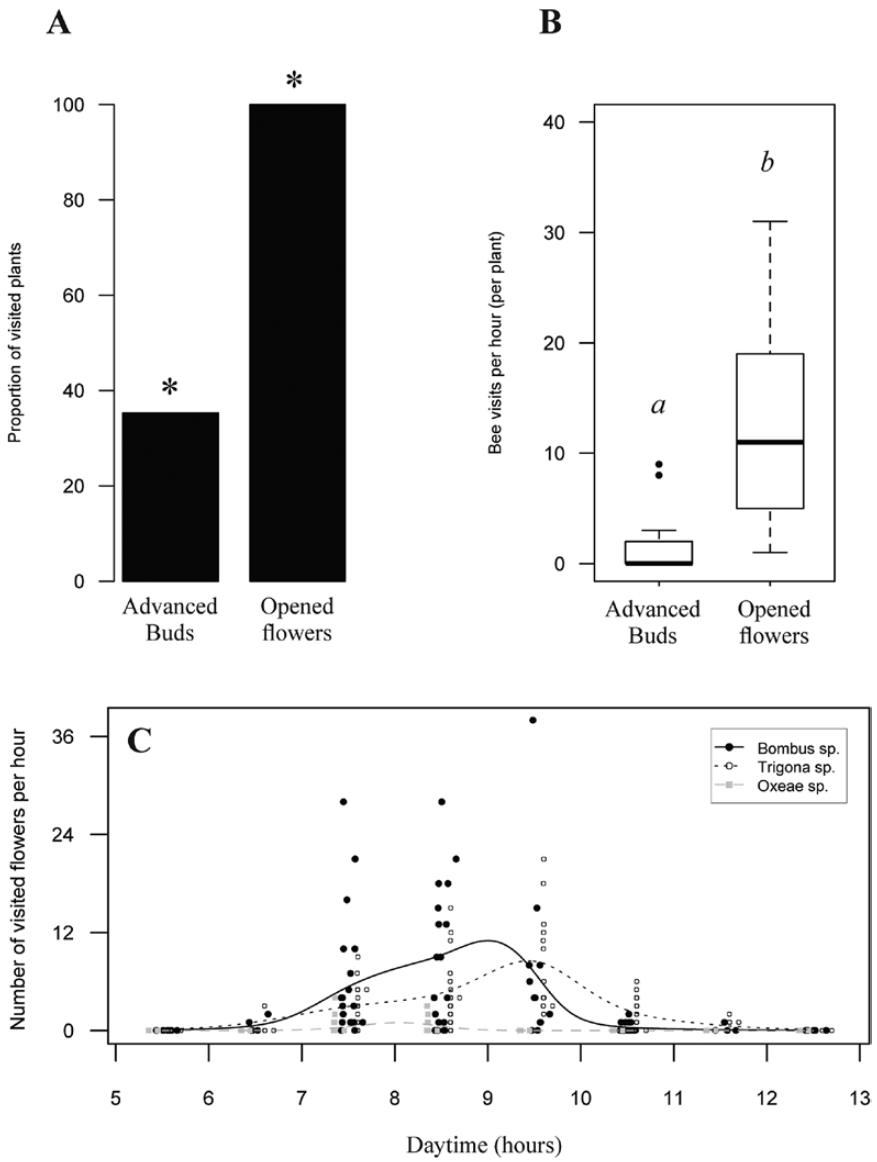


Figure 6. Frequency of bee visitors in anthetic flowers and pre-anthetic buds and pattern of bee visitation throughout floral anthesis in *C. desvauxii* var. *latistipula*. A, proportion of plants visited by bees exploiting pollen grains during a period in which bees visited all plants with anthetic flowers and only some plants with pre-anthetic buds. Asterisks indicate statistical differences between proportions ($P < 0.05$). B, on average, bee visits exploited four times more opened flowers than pre-anthetic buds. Different letters indicate statistical differences between the average number of bee visits between floral stages ($P < 0.05$). C, temporal pattern of bee species collecting pollen grains in anthetic flowers.

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11:30 h, with a peak during the first 3 h in the morning (Fig. 6C). *Bombus* sp. exploited flowers throughout the lifespan of the flower, with a peak around 09:00 h, showing a higher visitation rate early in the morning, when compared with other floral visitors (black line in Fig. 6C; $Z = 61.0$, $P < 0.001$, $N = 90$). *Trigona* sp. also visited flowers throughout the lifespan of the flower, with a peak around 09:30 h (dashed line in Fig. 6C; $Z = 84.0$, $P < 0.001$, $N = 90$), whereas *Oxaea* sp. was relatively rare throughout anthesis (grey line in Fig. 6C; $Z = 1.7$, $P = 0.69$, $N = 90$). Detailed statistical results can be found in Supporting Information Table S3. The two additional floral visitors, *Apis mellifera* and an unidentified small bee, were extremely rare and corresponded to < 3% of the total number of visits to *C. desvauxii* flowers. For *A. mellifera* (*c*. 2%), bees probably exploited pollen grains scattered on the cucullus after vibration by *Bombus* sp., as *A. mellifera* does not perform sonication.

We identified three distinct types of pollen-gathering exploitative behaviour among these bee species: (1) vibration of anthers on opened flowers (*Bombus* sp. and *Oxaea* sp.; Fig. 1D); (2) slow slicing of anthers on opened flowers, as well as petals in pre-anthetic floral buds, performed by bees using their mandibles (*Trigona* sp.; Fig. 1G–I); and (3) forcefully opening pre-anthetic floral buds with subsequent vibration of anthers (*Bombus* sp.; Fig. 1E, F). On average, *Bombus* sp. vibrated anthers in six flowers per plant per hour; *Oxaea* sp. exhibited the same behaviour, but on average in less than one flower per plant per hour (Table 2). During vibration of opened flowers, both species usually touched the stigma, making them potential pollinators of *C. desvauxii*. *Bombus* sp. was seen grabbing the base of long anthers with the mandibles while buzzing the whole set of anthers, generally causing small injuries at the base of long anthers (Fig. 1C, red arrows). The body of these bees became covered by pollen grains after sonicating anthers and the inner side of the cucullus (Fig. 1C). Bees collected part of the pollen grains during their release from poricidal anthers, but the portion of pollen grains deposited on bee flanks was not easily collected and remained spread on the body of the bee. *Bombus* sp. also had an alternative behaviour of flower visitation, i.e. forcing entrance to pre-anthetic buds and then vibrating the anthers (Fig. 1E, F). *Trigona* sp. acted as a floral antagonist. In anthetic flowers, these bees bit and cut the anthers with their mandibles for pollen collection, rarely touching the stigma. In addition, *Trigona* sp. damaged floral tissues, especially petals, in order to access the anthers in pre-anthetic buds and then performed the same behaviour on anthers of anthetic flowers (Fig. 1H, I). *Trigona* sp. exploited pollen in four flowers per hour per plant (Table 2).

Manual pollination experiments

Among the three manipulative pollination treatments, we found a high rate of fruit formation between all types of crosses. Following treatments in which anthetic flowers and pre-anthetic buds received pollen grains from recently opened flowers, all flowers formed fruits (100%), whereas crosses between pre-anthetic buds had 80% of buds forming fruits. Finally, pre-anthetic buds were completely isolated from floral visitors in the fourth treatment testing autonomous self-pollination; here, flowers set no fruit (0%).

DISCUSSION

Here we have described floral development in *C. desvauxii* var. *latistipula* with a particular focus on the features of pre-anthetic buds and anthetic flowers, stages during which bee visitors can access pollen grains as a food resource. In pre-anthetic buds, when sepals no longer protect inner whorls, petals expand in an exponential pattern, protecting the reproductive

Table 2. Behavioural descriptors of bee visitors on anthetic flowers of *Chamaecrista desvauxii* var. *latistipula* (Q = quartiles)

Bee visitors	Number of visited flowers per plant (1 h)			Number of buzzes per plant (1 h)			Contact with reproductive structures		Potential pollinator?
	Median (range)	1Q	3Q	Median (range)	1Q	3Q	Anthers (behaviour)	Stigma	
Bombus sp.	6 (0–38)	1	13	10 (0–64)	3	18	Yes (vibration)	Yes	Yes
Oxeae sp.	0 (0–4)	0	<1	0 (0–4)	0	1	Yes (vibration)	Yes	Yes
Trigona sp.	4 (0–21)	3	7	–	–	–	Yes (slicing)	Eventually	No
Apis sp.	0 (0–16)	0	<1	–	–	–	No	Eventually	No

The number of visited flowers and buzzes were quantified on an hourly basis, considering peak interval of bee visitation from 07:00 to 10:00 h.

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floral organs. However, petal length is characterized by a huge variability among pre-anthetic buds, allowing bees to exploit anthers in some pre-anthetic buds by using different strategies. In contrast, anthers grow linearly from young bud stages to flower opening, and some of them are fully developed and dehiscent in pre-anthetic floral buds. At this phase, when some anther pores are opened, pollen grains are still available for sonication. Pollen grains from long and short anthers are equal in size and have similar histochemical features, suggesting that stamen differentiation into pollinating and feeding anthers, in the context of the 'division-of-labour' hypothesis, is not accompanied by structural or functional differences in pollen grains of this species.

The three most common bee visitors accessed pollen grains in anthetic flowers and before flower opening (pre-anthetic buds) in *C. desvauxii* var. *latistipula*, although most visits occurred during anthesis. *Bombus* sp. and *Trigona* sp. were the most common pollen-collecting visitors, either by vibrating the anthers and touching reproductive organs (*Bombus*; potential pollinator), or by slicing anthers with mandibles without touching reproductive organs [*Trigona*; pollen robber *sensu* Inouye (1980)]. *Bombus* sp. was the first bee to access flowers and pollen during flower anthesis in the morning, potentially decreasing the cost of pollen grains otherwise lost to exploitation by non-pollinators. Thus, if most pollen robbers make their visits after pollinators have already made their collections, then they should not be considered as having a significant effect on the fruit set, as observed by Quinalha *et al.* (2017) for other savanna shrub species.

Pollen grains have the highest germination rate at the pre-anthetic bud stage and in anthetic flowers, and the same pattern was observed for stigma receptivity. Therefore, although floral whorls are neither completely developed at pre-anthesis stage in some floral units nor finally positioned, thereby reducing the chances of pollination (Almeida *et al.*, 2013b), it is still possible that pollination occurs, considering that *Bombus* bees buzz them. This possibility is reinforced by our finding that artificial pollinations at the pre-anthetic bud stage produced fruits in almost all manual crosses in spite of pollen donors being pre-anthetic buds or anthetic flowers.

Functional floral morphology

Buzz pollination is usually reported for other species in all genera of Cassiinae, including *Cassia* and *Senna*, and other *Chamaecrista* spp. (e.g. Buchmann, 1974; Delgado-Salinas & Sousa-Sánchez, 1977; Dulberger, 1981; Gottsberger & Silberbauer-Gottsberger, 1988; Hardin *et al.*, 1972; Delgado-Salinas *et al.*, 2011; Venkatesh, 1956a, b, 1957; Wolfe & Estes, 1992). In general, flower size of buzz-pollinated species was previously assumed to be related to the size of bee pollinators (see Solís-Montero & Vallejo-Marín, 2017). This phenomenon can be explained by the morphological fit between pollinator body size and the distance between the stigma and pollination anthers, which favours a particular bee touching the stigma (Willmer, 2011; Solís-Montero & Vallejo-Marín, 2017). Indeed, *C. desvauxii* varieties with larger flowers, like our focal variety, would be most effectively pollinated by large- and medium-sized bees, such as *Bombus* and *Oxaea*, which often touched the stigma during sonication, thus performing pollination behaviour. Other bee species with smaller body sizes are reportedly capable of visiting and pollinating large-flowered *Chamaecrista* spp., but they were considered less effective pollinators in these cases (Gottsberger & Silberbauer-Gottsberger, 1988). However, because we did not measure pollination efficiency, this is still a matter to be explored in future studies.

The large innermost lateral concave petal of *C. desvauxii* (the cucullus) was already reported for other *Chamaecrista* spp. (e.g. by Wolfe & Estes, 1992), but with different shape, position and function. Here, we observed that the cucullus is slightly thicker and more robust than other petals. This structure is also found in other species of the genus and in *Senna* spp. (Gottsberger & Silberbauer-Gottsberger 1988; Marazzi & Endress, 2008). In *C. desvauxii* var. *latistipula*, we observed that the cucullus may retain a large number of pollen grains on its inner side after sonication by bees. During sonication behaviour, bees become closely positioned to the inner side of the cucullus, especially the most frequent bee visitor, *Bombus* sp. Thus, after the first bee visit and sonication, other bees can receive a load of pollen grains in subsequent visits just by touching that petal while landing in the inner part of the flower (herein after termed as 'landing pollen receipt'). On that basis, we may consider that the cucullus not only 'ricochets' pollen (a direct outcome of vibration; Westerkamp, 2004; this study), but could also make pollen grains indirectly available by simply keeping part of the pollen on its surface. As this residual pollen is not collected during the first bee vibration, it becomes available for pollination in successive visits. In both cases ('ricochet' and 'landing'), the cucullus plays a crucial role in mediating pollen transfer, as similarly assumed in other studies with Cassiinae (Delgado-Salinas & Sousa-Sánchez, 1977; Westerkamp, 2004; Amorin *et al.*, 2017). Indeed, an exclusion experiment that removed the cucullus in flowers of *C. fasciculata*, a species with floral structure similar to that of *Chamaecrista desvauxii*, showed that manipulated flowers without the cucullus produced

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nine times fewer fruits than control flowers that kept this modified petal (Wolfe & Estes, 1992).

The heterantherous androecium in *C. desvauxii* var. *latistipula* shows no shape or size differences in pollen grains produced by either type of anther. We also did not detect any difference in the cytoplasmic content or the proportion of germinated pollen grains between these anthers, a finding contrary to our expectations based on recent data from species of Faboideae (=Papilionoideae) (e.g. Paulino, Mansano & Teixeira, 2013; Paulino *et al.*, 2016). Consequently, in our study, we did not find any evidence for the 'division-of-labour' hypothesis based on qualitative differences in the pollen grains, i.e. no clear differentiation between pollen from 'feeding stamens', which is exploited for the larval provision of bees, and pollen from 'pollinating stamens', which is available for pollination (Müller, 1881, 1882, 1883). Additionally, in pollen of both anther types, we verified the occurrence of proteins and fatty acids, compounds of high nutritional value and congruent with the use of pollen grains as a food resource (Edlund, Swanson & Preuss, 2004). The occurrence of intracellular phenolic compounds may be important for bees, owing to their antioxidant and antimicrobial effects (Ferreyra, Rius & Casati, 2012). Moreover, from an ecological point of view, pollen fluorescence emission in a shorter wavelength region (UV) (Mukherjee & Chanda, 1990) may be a signal to insects searching for the resource location on the anther tips inside the flower. Therefore, pollen from heterantherous flowers of *C. desvauxii* var. *latistipula* shows features that might attract bees looking for food supply, irrespective of whether it comes from short or long anthers. Detailed studies comparing the morphology and function of pollen grains in heteromorphic stamens have rarely been conducted in Fabaceae, but see Luo *et al.* (2009) and Paulino *et al.* (2013, 2016) for examples.

On the other hand, the morphological arrangement of stamens may represent a good adjustment for pollination, corroborating of 'division-of-labour' hypothesis in the focal species. We verified that the three long anthers varied their position in flowers, but always facing the inner side of the cucullus and with a length similar to that of the carpel. Furthermore, compared to short anthers, long anthers had more pollen grains. Thus, in our study, we found that the potential difference in the role of anthers with variable lengths could not be associated with the pollen itself, but instead with the position of anthers relative to other whorls, especially the cucullus, and with the amount of pollen grains available for transfer. In contrast, some species of Faboideae with heterantherous androecia, such as *Swartzia dipetala* Willd. ex Vogel, *Lupinus* 'The Governor' and *Cytisus scoparius*, have shown some differences in pollen grains in favour of the 'division-of-labour' hypothesis (Paulino *et al.*, 2013, 2016). For *C. fasciculata*, Wolfe & Estes (1992) found no differences in pollen grain germination in relation to heteranthery, similar to our findings. For *C. desvauxii*, a morphological size adjustment of the three long anthers and carpel could take place, each one positioned sideways in each flower. If long anthers are able to place pollen grains in a specific spot on the body of a bee such that pollination is favoured, in a manner different from that of short anthers, then we could advocate the 'division-of-labour' hypothesis in *Chamaecrista desvauxii*.

Chamaecrista desvauxii var. *latistipula* has enantiostylous, monomorphic flowers with the carpel deflected in the lower part of the flower, but always positioned sideways to the cucullus and anthers, with the stigma facing the inner side of the cucullus. This suite of features may have an important impact on pollen transfer to stigmas, because pollen grains from each flower morph will be deposited on opposite flanks of the body of the bee, which leads to pollen transfer from one morph to another, preventing intramorph pollination (Barrett, 2002; Almeida *et al.*, 2015). This pattern may reduce the probability of geitonogamy in self-compatible plants, especially in smaller plants, which usually have few opened flowers per day and only one floral morph available. However, enantiostyly cannot avoid geitonogamy completely because larger plants in the population have flowers of both left- and right-morphs daily in the same plant.

Stigma structure found in *C. desvauxii* mostly follows previous morphological studies (Owens, 1989). A crateriform stigma, with a porate entrance lined with a fringe of hairs, was previously described for other species in this genus and for some *Senna* spp. (Owens, 1989; Tucker, 1996; Marazzi *et al.*, 2007). However, in previous floral biology studies, the stigmatic pore in *Chamaecrista* spp. was described as usually closed by the stigmatic hairs in its entrance, based solely on SEM analysis. Here we show that the stigmatic entrance is always open in buds and fresh anthetic flowers of *C. desvauxii* var. *latistipula* and that the closed state is an artefact of SEM procedures. Therefore, previous studies based solely on SEM analysis may have mistakenly described the closed state of the stigmatic entrance of *Chamaecrista* spp. Another important feature we have found is the presence of exudates on the stigma, which is different from other species of Cassiinae thus far studied (see Costa *et al.*, 2014).

Timing of flower whorl development and bee visitation pattern

The developmental studies of Tucker with flowers of Cassiinae, like those with *Chamaecrista* (Tucker, 1996, 1997), considered the timing of floral organ development only in an indirect fashion. That is,

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Tucker inferred bud age using size as an indirect parameter for developmental time. Our study is one of the first to directly associate floral organ development with floral biology and pollination.

In many angiosperm species, sepals are the protective organs of inner floral whorls in the bud stage (Weberling, 1989; Endress, 1994). In fact, during floral development, especially in young buds, the calyx acts as a protective whorl, and petal development is delayed in relation to the androecium at this stage. However, petals assume the protective role of anthers and carpel in the last stages of floral development in *C. desvauxii* var. *latistipula*. A similar delay in corolla development after initiation was observed for other eudicots and monocots that experience intense growth, but only close to flower opening (Hiepko, 1965; Leins & Stadler, 1973; Endress, 2011), probably as a reflection of the exponential pattern and high interfloral variability found here for pre-anthetic buds of *C. desvauxii.* Corolla size variation would determine the pattern of petal overlap in pre-anthetic buds, a possible determinant of pollen exploitation by bees before flower opening because buds can become easier to open when the corolla is not fully grown. Pollen exploitation before anthesis may have two important consequences for the sexual reproduction of this species. First, if bee behaviour allows pollen transfer to stigma, the flower may be pollinated at the bud stage, thus improving the reproductive success of the plant. Second, if these bees act as pollen thieves, like *Bombus* that vibrate the anthers, or like *Trigona* that acts as a pollen robber, damaging the anthers, considerable pollen waste could occur before anthesis, which would probably have a negative impact on plant fitness. The success of *Bombus* buzz pollination vs. the impact of *Trigona* exploitation on pre-anthetic bud stage remains to be investigated in future reproductive studies.

The anthers differing in morphology and developmental chronology produce similar pollen grains, which are available at stages prior to anthesis. This may explain the occurrence of bee visits in pre-anthetic floral buds in *C. desvauxii* var. *latistipula*, given that a highly nutritious pollen resource is available at this stage. Poricidal anthers could be considered as a floral trait that makes pollen access difficult for non-buzzing bee visitors (Gottsberger & Silberbauer-Gottsberger, 1988). However, in our study, we verified that small bees could cut the anthers and get the pollen in an illegitimate way, including at flower bud stages. Additionally, in pre-anthetic buds, we verified that pollen grains are viable, anther pores are partially open and the stigma is receptive. These features could allow flowers to be pollinated prior to flower opening, even without having all whorls fully developed. In fact, in the pre-anthetic bud stage, the cucullus encloses not only all stamens, but also the gynoecium of *C. desvauxii*, which could favour self-pollination during sonication at this floral stage, representing an alternative pollination mode.

In conclusion, this study highlights the existence of an extended window of opportunity for mutualistic interactions between bees and flowers at different development stages of a typical pollen-flower species. Floral attractiveness, commonly attributed to the fully developed floral whorls and pollinator sensory abilities, was apparently lengthened by the resource supply at pre-anthesis stages. Indeed, in some way, bees were attracted to floral buds, and their behavioural alternatives allowed them to exploit pollen grains. Thus, pollination may occur at pre-anthetic buds in our focal species, even though the buds have a distinct shape and arrangement of reproductive and sterile whorls when compared to anthetic flowers. In summary, variation in floral morphogenesis, flower functional features and bee-pollinator behaviours over time constitute an essential aspect of pollination in this plant species and could favour the appearance of new modes of interaction, ultimately changing the evolutionary reproductive pathway of these organisms.

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Table S1. Detailed statistical results and modelling used in the graphs of Figure 3.
Table S2. Detailed statistical results and modelling used in the graphs of Figure 5.
Table S3. Detailed statistical results and modelling used in the graphs of Figure 6.

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